

Examiner reports

Q1.

A very small number of candidates seemed unsure how to deal with the vectors and made very little progress with the question. The majority of candidates did produce quite good solutions. Part (a) did not cause any particular difficulties. In part (b), some

candidates wrote $\sqrt{9^2 - 4^2}$ instead of $\sqrt{9^2 + 4^2}$. Those who could visualise part (c) had no difficulty, but some found it hard to identify a triangle to work in and tried to use the sine or cosine rules. A few candidates produced incorrect expressions involving sin or cos instead of using tan.

Q2.

Many of the candidates found this question more difficult than expected. Part (a) was generally done well, but problems then emerged in part (b). Here many candidates found the magnitude of the acceleration and then used this scalar quantity in part (c). It was not uncommon to see an equation that involved a velocity that was a vector and an acceleration that was a scalar. Some of the candidates who worked through correctly stopped in part (c) and did not find the distance that had been requested.

Q3.

This question was quite challenging for the candidates and there were very few good solutions, although most candidates did pick up marks from one or two parts of the question. Part (a) was done by a fair number of candidates. Errors included treating the position at $t = 10$ as a velocity and not using this position vector at all. There were some good attempts at part (b) with candidates using their acceleration from part (a). Part (c) caused problems for many candidates. It was not uncommon to see answers where the time was given as a vector. Some candidates only considered one of the components. There was clearly a number of candidates who could write down a vector equation, but could not solve it to find T . In part (c)(ii), several candidates only found the velocity and did not go on to find the speed.

©2025 Exam Papers Practice. All rights reserved.